

ZAYNAB EL HAKOUR

zaynab.elhakour@yotes.collegeofidaho.edu | (986) 837-7683 | <https://github.com/zaynabhkr>

Education

The College of Idaho (GPA: 3.80)

Aug 2022 - May. 2026

Bachelor of Arts in Computer Science and Mathematics

Caldwell, ID

- Minors: Art and Design, Political Economy
- Gipson Honors Student; Kathryn Albertson Scholar; SWE Vacasa Scholar; Dean's List; RTC Member; WiCyS Member

Technical Skills

- **Languages:** Python (Proficient), SQL (PostgreSQL, MySQL), C++, TypeScript
- **Data & Databases:** MySQL, MongoDB, relational schema design, query optimization
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, RDS) Cloud Practitioner Certified, Docker, GitHub Actions, CI/CD pipelines, Jenkins
- **Backend & APIs:** Node.js, Express.js, REST APIs, FastAPI
- **Tools:** Git/GitHub, JIRA, VS Code, Linux/UNIX CLI, Makefile, CMake

Work Experience

Software Engineering Intern

Aug. 2025 - Present

Schweitzer Engineering Laboratories

Boise, ID

- Execute requirements-based functional tests on SEL-2731 and SEL-2741 Ethernet switches by configuring relays, power systems, and networks to ensure ISO-compliant reliability, preventing costly field issues for utility clients.
- Develop automated test scripts using Python and Gherkin (BDD) frameworks to replicate and streamline manual test cases, reducing manual testing time by 40% and ensuring consistent validation across regression and integration cycles.
- Utilize UNIX/Linux and Windows environments to install dependencies, manage builds, and troubleshoot device configurations, integrating test results within SEL's CI/CD pipeline.
- Maintain 100 % requirement-to-defect traceability through JIRA and proprietary tools, enabling managers, developers, and clients to verify product readiness prior to release.

Software Developer Intern

Jun. 2025 – Aug. 2025

SWCA Environmental Consultants

Phoenix, AZ

- Developed backend services using Python and FastAPI, applying object-oriented programming to design modular API endpoints and routing logic
- Built data ingestion workflows through webhook-based systems to process and update structured data from ArcGIS Feature Layers
- Designed and maintained RESTful APIs to support data extraction, transformation, and CRUD operations
- Automated build and deployment processes using Makefiles and Docker, supporting CI/CD workflows from source code to containerized applications
- Collaborated with frontend developers to integrate APIs with React.js interfaces, improving data accessibility and usability through optimized table components

Computer Science Teacher Assistant

Aug. 2023 – Jun. 2024

The MAPS Department, The College of Idaho

Caldwell, ID

- Guided over 40 students through core programming principles, with a strong emphasis on Python best practices, including syntax precision, code readability, and debugging techniques.

Leadership/Volunteer

President, International Students Organization (ISO)

May 2025 – May 2026

- Lead the largest cultural organization on campus and coordinated events with 100 participants to foster inclusion and global awareness.

Co-President and Co-Founder, Coding Club

Sep. 2022 – Jun. 2024

- Organized and coached weekly LeetCode coding nights for 25 members to prepare undergraduate fellows for internship interviews.

Technical Project

Granfield College Database System (SQL, Relational Design, UML)

- Designed and implemented a relational database system for a simulated college environment, translating raw institutional data about software licenses into structured schemas
- Modeled entity relationships using UML diagrams (one-to-one, one-to-many, many-to-many) to ensure data integrity and scalability
- Defined data constraints, normalization rules, and stakeholder requirements to maintain consistency and prevent redundancy
- Wrote complex SQL queries (joins, aggregations, subqueries) to extract and analyze data for reporting and decision-making

Awards

- First Female Prize in the 2020 Junior Olympiad in Programming in C++.
- Model United Nations NYC 2024 Outstanding Delegation Award: Sri Lanka.
- Certification of Global Citizenship from Legacy International and the U.S. Bureau of ECA for 2020 TechGirls Program Leadership.

Languages

English, Arabic, French.